

SAMUEL MICHAELS RODRIGUEZ

Software Engineer | Generative AI & Data Engineering

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PROFESSIONAL SUMMARY

Bachelor of Science honours graduate in Computer Science with a Mathematics minor (3.9 GPA, Dean's List) and two years of professional software and ML engineering experience, drawn to generative AI and data engineering. Ships full-stack production systems end to end: an LLM + RAG phishing-detection service, a containerized document-intelligence API covered by automated tests and continuous delivery, and a React web app delivered to a paying client. Comfortable in both application-development and research environments, with real experience modernizing legacy code — a monolithic C simulation restructured into clean, modular, tested units. Learns new languages fast, works in tight feedback loops, and genuinely enjoys collaborating and pairing with other engineers.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL, C, Bash, HTML/CSS

Generative AI: LLM APIs (Anthropic, OpenAI, Groq, Ollama), RAG pipelines, LangChain, ChromaDB & vector databases, embeddings, prompt engineering, LLM evaluation, agentic workflows

Data & ML Engineering: PyTorch, HuggingFace Transformers, scikit-learn, XGBoost, NumPy, pandas, ETL & data pipelines, schema design, PostgreSQL, Redis, SQLite

Full Stack Development: FastAPI, REST API design, React, Next.js, Vite, Tailwind CSS, Node.js, Streamlit — comfortable as a full stack developer across interface, service, and data layers

Engineering Practices: Test-driven development, pytest, refactoring, code review, pair programming, extreme programming (XP) practices, continuous integration and continuous delivery (GitHub Actions), agile iteration, mentorship, security-aware CI/CD (DevSecOps practices), Git, Docker, Linux

Deployment & Operations: Docker, Railway, Render, Vercel, HuggingFace Spaces, GitHub Pages, cloud deployment, production debugging

PROFESSIONAL EXPERIENCE

Machine Learning Research Developer

May 2025 – Apr 2026

Western University — London, ON

- Increased syllable-boundary detection accuracy from an F1 of 0.42 to 0.77 — an 83% relative gain, statistically validated at $p = 0.016$ (Cohen's $d = 2.61$) — by augmenting the VG-HuBERT transformer with a task-specific segmentation head in PyTorch.
- Built a reproducible evaluation framework across 6 subjects by engineering a leave-one-run-out cross-validation data pipeline in Python with time-aligned ground-truth labels at 30 ms tolerance.
- Kept a research codebase trustworthy and repeatable by version-controlling every training and evaluation run in Git on Linux, producing publication-quality analysis and figures.
- Operated in a research environment while shipping working software — the combination of application development and applied research this role asks for.

Software Developer (Research) — Legacy System Modernization

May 2024 – Sept 2024

Western University — London, ON

- Modernized a legacy C simulation (bacterial-growth modelling) by decomposing monolithic logic into clean, modular functions — eliminating 100% of detected memory leaks and removing extensive duplicated code.
- Improved runtime efficiency and long-term maintainability by restructuring and standardizing the codebase so the lab could safely extend it — the same discovery-then-safely-rewrite pattern Mechanical Orchard applies at scale.
- Conducted technical discovery on an unfamiliar codebase written by others, mapping behaviour before changing it.

Freelance Full-Stack Developer

2025 – 2026

Self-employed — London, ON

- Shipped a production multi-page web application (SigmaK9 Training) in React (Vite) with Tailwind CSS, client-side routing, and a filterable catalog of 12+ products across 4 categories, deployed to Vercel.
- Delivered on schedule and to spec by gathering requirements from a non-technical client and translating them into working features through short, agile feedback cycles.

Produce Supervisor

Sept 2020 – Present

Sunripe Fresh Market — London, ON

- Led and trained a 4–6 person crew across 5 years, driving shelf-tag scanning errors from recurring to zero — sustained while carrying a full-time honours course load.

PROJECTS

PhishGuard — AI Phishing Detection (Browser Extension + API) | *Python, FastAPI, LLM/RAG, ChromaDB, Redis, Docker* | github.com/sam-michaels/phishguard

- Built a cross-browser WebExtension (Chrome & Firefox, Manifest V3) backed by a FastAPI service that scores URLs in real time and returns an explainable risk verdict.
- Engineered a multi-signal detection pipeline combining heuristics, typosquatting detection, WHOIS domain-age checks, VirusTotal reputation scoring, and an LLM + RAG layer (ChromaDB over threat-intelligence sources) for context-grounded reasoning.
- Operated it as a production service: added a versioned Redis cache to cut latency and API cost, then diagnosed and fixed live issues including stale cache verdicts and false-positive scoring. Deployed via Railway.

RAG Document Intelligence API | *LangChain, ChromaDB, FastAPI, Docker, pytest* | github.com/sam-michaels/rag-document-api

- Built and deployed a retrieval-augmented document Q&A service: PDF ingestion, chunking, embedding, and semantic retrieval over a ChromaDB vector store behind a documented FastAPI REST API.
- Grounded LLM responses in retrieved context to prevent hallucination; containerized with Docker and covered with automated pytest tests written alongside the features, not after.

Personal Portfolio & Research Site | *Vanilla JS, HTML5/CSS3, Canvas API, DSP* | sam-michaels.github.io

- Designed, built, and actively maintain a zero-dependency portfolio site featuring an interactive demo of my speech-segmentation research: predicted vs. ground-truth boundaries, an adjustable matching tolerance, and precision/recall/F1 recomputed live in the browser.
- Implemented a short-time Fourier transform from scratch — hand-written radix-2 FFT, Hann windowing, dB magnitude mapping — to render a real spectrogram client-side, and deferred it off the critical path to keep first paint under 10 ms.
- Shipped light/dark theming with persisted user preference and a fully responsive single-file build deployed on GitHub Pages.

Connect Four Reinforcement Learning Agents | *PyTorch — DQN, PPO, CNNs (NeurIPS-format paper)* | github.com/sam-michaels/connect-four-rl

- Trained and compared DQN and PPO agents and authored a NeurIPS-format paper; novel contributions included supervised pretraining, a dual-path CNN, and a hybrid minimax-RL inference agent.

EDUCATION

Bachelor of Science (B.Sc.) Honours, Computer Science

June 2026

Western University — London, ON

Minor in Mathematics

3.9 GPA | Dean's List | Degree conferred June 2026 | Available to start immediately

Computer Science coursework: Machine Learning, Deep Learning, Artificial Intelligence, Reinforcement Learning, Operating Systems, Data Structures & Algorithms, Databases.

Mathematics coursework: Intermediate Linear Algebra, Advanced Calculus, Abstract Algebra, Discrete Mathematics, Calculus-Based Statistics and Probability.